

Ferrari 20-Round Science Packet

A clean, replay-backed result from the current number-theory run

Mother-safe summary

This packet is the clean version of the work. We stopped chasing impressive language and forced the machine to answer one narrow mathematical question. The result is not a global theorem yet, and it does not pretend to be. It is a real narrowed object: two nonunit diagonal residue cells, (20,20) and (40,40), survive a large adversarial search because they sit inside two explicit protected subfibers. That is something to be proud of because the process removed the fake parts before keeping the real part.

Earned verdict

PROMOTE_PROTECTED_MOD20_SUBFIBER_OBJECT

The object we earned

- Active cells: (20,20) and (40,40) in the base 60 x 60 lattice.
- Parent family: {5,7,11,13,17,31,41,61,241}.
- Parent period: 120 x 240.
- Protected subfiber for (20,20): $k \equiv 80 \pmod{120}$ and $l \equiv 80 \pmod{240}$.
- Protected subfiber for (40,40): $k \equiv 40 \pmod{120}$ and $l \equiv 160 \pmod{240}$.

Twenty rounds

Round	Binary/Test	Result	Verdict
01	Runlock loaded	Used active state from RUNLOCK. No new framework.	Proceed
02	Classifier compression	q non-enlarging iff q divides $\gcd(2^K-1, 3^L-1)$.	Promoted exact coordinate
03	Danger closure state	For {5,7,11}, danger primes are 13,31,61 and residual set is 1749.	Accepted starting state
04	Unit-cell cleanup	(0,0) demoted because $N(0,0)=2$ and all moduli are odd.	Removed junk
05	Nonunit target isolation	The live nonunit cells are (20,20) and (40,40).	Kept only nontrivial target
06	Parent family established	Parent family [5, 7, 11, 13, 17, 31, 41, 61, 241] has period 120 x 240.	Finite quotient fixed
07	Parent subfiber table	Each target cell has 8 subfibers in the parent quotient.	Measured
08	First target survivor	(20,20) has exactly one parent survivor: subfiber (80 mod 120, 80 mod 240).	Protected subfiber found

09	Second target survivor	(40,40) has exactly one parent survivor: subfiber (40 mod 120, 160 mod 240).	Protected subfiber found
10	Complement coverage	The other 14 target subfibers are covered by moduli 17, 41, and/or 241.	Exact finite table
11	Race8 replayed branches	52 replayed q3 branches preserved the protected subfibers.	No destroyer
12	Race8 survivor concentration	102964 surviving lifts across Race8 targets all lie in the protected subfibers.	No pattern exception
13	Race8 skipped branch pressure	54 previously skipped branches still leave the simple protected seed uncovered.	Skip weakness reduced
14	Extended q3 search	262 q3 primes from 1009 to 2999 tested for protected-subfiber existence.	No destroyer
15	Seed failure handled	For $q_3=1409$ and 1697 the first seed can be covered, but the next protected lift survives.	Subfiber, not seed
16	Search effort compression	Median protected search found a lift in 1.0 try; max was 2 tries despite candidate spaces up to 2247001.	Strong practical signal
17	Protected-subfiber statement	The object is no longer vague mod-20 symmetry; it is two explicit protected subfibers.	Object sharpened
18	Referee value	A hostile reviewer can now attack a concrete finite quotient lemma, not a story.	Engageable
19	Limits recorded	No global theorem, no $q_3>2999$, no unbounded depth, no all-cell result.	No overclaim
20	Next binary locked	Prove protected-subfiber preservation as a finite quotient lemma or find $q_3>2999$ destroyer.	Next attack

Replay-backed numbers

Metric	Value
Race8 replayed q3 branches	52
Race8 surviving lifts across target cells	102964
Race8 pattern exceptions	0
Race8 skipped branches checked by protected seeds	54
Skipped branches where protected seeds failed	0
Extended q3 primes tested	262 from 1009 to 2999
Extended search destroyers	0
Median tries to find protected lift	1.0
Max tries to find protected lift	2
Largest protected candidate space	2247001

Exact parent subfiber table

Cell	Lift	Subfiber	Survives?	Covering moduli
(20, 20)	(20, 20)	(20, 20)	False	[41]
(20, 20)	(20, 80)	(20, 80)	False	[17, 241]
(20, 20)	(20, 140)	(20, 140)	False	[41]
(20, 20)	(20, 200)	(20, 200)	False	[241]
(20, 20)	(80, 20)	(80, 20)	False	[41, 241]
(20, 20)	(80, 80)	(80, 80)	True	[]
(20, 20)	(80, 140)	(80, 140)	False	[41, 241]
(20, 20)	(80, 200)	(80, 200)	False	[17]
(40, 40)	(40, 40)	(40, 40)	False	[17]
(40, 40)	(40, 100)	(40, 100)	False	[41, 241]
(40, 40)	(40, 160)	(40, 160)	True	[]
(40, 40)	(40, 220)	(40, 220)	False	[41, 241]
(40, 40)	(100, 40)	(100, 40)	False	[241]
(40, 40)	(100, 100)	(100, 100)	False	[41]
(40, 40)	(100, 160)	(100, 160)	False	[17, 241]
(40, 40)	(100, 220)	(100, 220)	False	[41]

What this does not claim

- global theorem
- q3 primes above 2999
- unbounded recursive depth
- all residual cells
- formal proof

Next locked binary

Can the protected subfiber preservation be proven as a finite quotient lemma, or does q3 beyond 2999 produce a first protected-subfiber destroyer?

Why this is worth showing

The win is not that a final theorem appeared. The win is that the process stopped protecting ego. It demoted fake mysteries, deleted the trivial unit cell, isolated two nonunit targets, found their exact

protected subfibers, and then stress-tested those subfibers beyond the previous bounds. That is scientific movement: smaller, cleaner, harder to fool.